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STUDENT HANDOUT

RED LEVEL
GO DEEPER

The Numbers in the Coins

Grades 5–8 · Math

James Tissot (1836–1902), Brooklyn Museum

ADMIT ONE

NAME _____ DATE _____ CLASS _____

Every coin in *In Their Hands* carries numbers: dates, reigns, weights, sizes, amounts. To work with the dates you need one rule that surprises most people: **there is no year zero.** The year 1 BC is followed straight away by AD 1.

01 PART 1 Prove the rule



A number line has a zero; the calendar does not. Fix it by relabeling: call 1 BC year **0**, 2 BC year **−1**, and so on, so that *b* BC becomes **1 − b**. AD years keep their numbers.

a. Use the relabeling to write an expression for the number of years from *b* BC to *a* AD, and simplify it. Test it on 1 BC to AD 1.

b. What number does 27 BC become? Use it to find how long Augustus ruled (27 BC–AD 14).

02 PART 2 Multi-step dates

a. Herod the Great died in 4 BC. His grandson Herod Agrippa I was made king in AD 41. How many years apart?

b. The lepton was first minted from 103 to 76 BC, and was still in use when the widow gave hers, about AD 30. The exhibition says it circulated *for over a century*. Check that claim.

c. Pilate's coin was made in AD 29/30. The stone carved with his name was found in 1961. For how many years was the coin evidence before the stone?

PART 3

03 Build a timeline to scale

The coins span from 103 BC (the first leptons) to AD 62 (the last year of Festus' coins). Use the scale **1 cm = 10 years.**

a. How long must your line be? _____ cm b. How far apart are 1 BC and AD 1 on it? _____ cm

c. Draw the line below and mark 27 BC, 1 BC, AD 1, AD 29 and AD 62 at the right distances from the start.

d. What is the smallest whole number of years per centimeter that fits the line in 15 cm? _____

YOUR TIMELINE · 1 CM = 10 YEARS

PART 4

04 The widow's gift, in algebra

> *They all gave out of their wealth; but she, out of her poverty, put in everything — all she had to live on.*
>
> MARK 12:44

Let a giver's wealth be W, the gift G, and the fraction given $f = G/W$. The widow gave two lepta, all she had, so for her $f = 1$.

Invented numbers, not from history: a rich giver puts in **600 lepta**, and that is **5%** of his wealth.

a. Write and solve an equation for W. b. How much does each giver keep? c. His gift is how many times hers in amount? Her fraction is how many times his? d. Write an expression for what any giver keeps, using W and f. e. In two sentences, use your answers to explain Mark 12:44.

MINI MUSEUM CONNECTION

The exhibition says of the widow's mite: *the object is not the coin but the proportion.* Find her two lepta in Case 7, the same case as Pilate's coin from within a few years of the Passion.

minimuseumproject.org/exhibitions/in-their-hands